

Guideline/Protocol Title	Prophylactic Antibiotics for Traumatic Open Fracture in Pediatrics
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Committee Review	Antimicrobial Stewardship Subcommittee Pharmacy and Therapeutics Committee
Target Population	Pediatric patients with traumatic open fractures being seen for initial fracture management.
Overview	This guideline provides clinicians with protocols for prophylactic antibiotic management as well as pertinent vaccination administration in the setting of traumatic open fractures.
Effective Date	10/2021
Revised Date	11/2024
Expiration Date	11/2027
Schedule for Periodic Review	Every 3 years
Implementation Strategy	See below. Each service will be trained on the guideline.
Education Strategy	Guideline will be posted to CareWeb under Antimicrobial Stewardship orders/protocols, added to the Antimicrobial Stewardship Application, and the Trauma Care Guidelines (Pediatric)
Primary Outcome(s)	Standardize antimicrobial prophylaxis for pediatric open fractures.
Outcome Assessment Plan	
Document Posting Location	☒ CareWeb Guidelines and Protocols ☐ Markey Cancer Center (MCC) Protocols ☒ Other Location: Antimicrobial Stewardship site
Information Technology Needs	☒ Link to LexiDrugs – Medication(s): clindamycin, doxycycline, levofloxacin, cefazolin, penicillin G, ampicillin/sulbactam, ceftriaxone, PCV-20, PCV13

PROPHYLACTIC ANTIBIOTICS FOR TRAUMATIC OPEN FRACTURE IN PEDIATRICS

Antimicrobial prophylaxis should be administered in the trauma bay **as soon as possible, ideally within 1 hour of injury**. Doses and frequencies listed detail/describe open fracture prophylaxis while awaiting closure. For antibiotic dosing prior to any operative incision and intraoperative dosing frequency, see [surgical prophylaxis guidelines](#) on CareWeb. Contact pharmacy with dosing questions.

Fracture or Wound Type (Gustilo classification)	Suggested Regimen <i>Initiate ASAP and within 1 hour of arrival to UK</i>		DURATION
	Primary	Type I (Anaphylactic) Penicillin Allergy	
Type I, II, or III Blunt or Penetrating^{1, 2} (including GSW)	Cefazolin 30 mg/kg IV Q8H Maximum per dose 2000 mg (if >120 kg max 3000 mg)	Clindamycin 10 mg/kg IV Q6H Maximum per dose 900 mg	Until soft-tissue coverage, and <u>no longer than 24</u> hours from injury
Soil or Fecal Contamination Type I, II, or III^{1, 2} Farm-related injury or fecal contamination	Cefazolin 30 mg/kg IV Q8H Maximum per dose 2000 mg (if >120 kg max 3000 mg) PLUS Penicillin G Potassium 66,667 units/kg IV Q4H x 6 doses Maximum per dose 4 million units	*Clindamycin 10 mg/kg IV Q6H Maximum per dose 900 mg	Until soft-tissue coverage, and <u>no longer than 24</u> hours from injury
Dog or Cat Bite Leading to Open Fracture¹⁻³	Ampicillin/Sulbactam 75 mg/kg IV Q6H Maximum per dose ampicillin 2000 mg <i>May consider de-escalation to amoxicillin/clavulanate 20 mg/kg/dose (max: 500 mg) TID following 24 hours of IV ampicillin/sulbactam</i>	Doxycycline 2.2 mg/kg IV/PO Q12H Maximum per dose 100 mg	3 days*
FRESH Water Injury Leading to Open Fracture^{1, 2}	Levofloxacin 10 mg/kg IV/PO Maximum per dose 750 mg PLUS Graded Fracture Antibiotics	Levofloxacin 10 mg/kg IV/PO Maximum per dose 750 mg PLUS Graded Fracture Antibiotics	Until soft-tissue coverage, and <u>no longer than 24</u> hours from injury
SALT Water Injury Leading to Open Fracture^{1, 2}	Doxycycline 2.2 mg/kg IV/PO Q12H Maximum per dose 100 mg PLUS Ceftriaxone 50 mg/kg IV Q24H Maximum per dose 2000 mg	CONSULT Pediatric AMS [#]	Until soft-tissue coverage, and <u>no longer than 24</u> hours from injury
Open Mandible[‡] or Open Facial Fracture⁴	NONE	NONE	-
Open Skull Fracture^{1, 2, 5}	NONE unless grossly contaminated <i>If grossly contaminated, prophylaxis as above for soil or fecal contamination</i>	NONE unless grossly contaminated <i>If grossly contaminated, prophylaxis as above for soil or fecal contamination</i>	Until soft-tissue coverage, and <u>no longer than 24</u> hours from injury
CSF Leak, Closed Facial Fracture, Sinus, Anterior, or Posterior Table^{1, 2, 5}	NONE PLUS Pneumococcal Vaccination Prior to Discharge [‡]	NONE PLUS Pneumococcal Vaccination Prior to Discharge [‡]	-

Antibiotics should be continued until soft-tissue coverage, and no longer than 24 hours total from injury, regardless of when antibiotics are initiated.

Fluoroquinolones are NOT recommended outside of fresh water injuries. Patients with isolated hand injuries or GSW to foot or transabdominal pathway into Hip/Pelvis/Spine should receive routine graded antibiotic prophylaxis ^{1, 2, 6, 7}; [‡]PLUS chlorhexidine rinses PO QID for mandible fractures; [‡] See Table III;

*Duration of 3 days as recommended for routine bite prophylaxis in IDSA SSTI guidelines; consider rabies vaccination and immunoglobulin;

[#]: Pediatric AMS may be reached via Epic secure chat (Katie Olney, PharmD and Joel Howard, MD)

Table IA: Gustilo Open Fracture Classification

Gustilo Grade	Definition
I	Clean wound <1 cm in length
II	Wound > 1 cm without extensive soft-tissue damage, flaps, avulsions
III	Extensive soft-tissue laceration, damage, or loss or an open segmental fracture.
IIIA	Adequate periosteal coverage despite the extensive soft-tissue laceration or damage
IIIB	Extensive soft-tissue loss and periosteal stripping and bone damage. Usually associated with massive contamination. Will often need further soft-tissue coverage procedure (i.e. free or rotational flap)
IIIC	Arterial injury requiring repair, irrespective of degree of soft-tissue injury

Table II: Tetanus Recommendations

History of Tetanus Vaccination	Last Dose	Wound Type	DTaP, Tdap or Td†	TIG‡
Unknown or <3 doses		Clean, minor wound	Yes	No
		All other wounds*	Yes	Yes
≥3 doses	<5 years	All wounds	No	No
	5-10 years	Clean, minor wounds	No	No
	5-10 years	All other wounds*	Yes	No
	≥10 years	All wounds	Yes	No

Abbreviations: DTaP = Diphtheria and Tetanus toxoids and acellular pertussis vaccine; Tdap = tetanus toxoid, reduced diphtheria toxoid, and acellular pertussis; Td = tetanus and diphtheria toxoids; TIG = Tetanus immune globulin

*Such as, but not limited to, wounds contaminated with dirt, feces, soil, and saliva; puncture wounds; avulsions; and wounds resulting from missiles, crushing, burns, and frostbite; † DTaP is recommended for children <7 years of age. Tdap is preferred to Td for persons aged 11 years or older who have not previously received Tdap. Persons aged 7 years or older who are not fully immunized against pertussis, tetanus, or diphtheria should receive one dose of Tdap (preferably the first) for wound management and as part of the catch-up series; if additional tetanus toxoid-containing doses are required, either Td or Tdap vaccine can be used; ‡ People with HIV infection or severe immunodeficiency who have contaminated wounds (including minor wounds) should also receive TIG, regardless of their history of tetanus immunizations.

Table III: Pneumococcal Vaccination in Patients with CSF Leak

Age <2 years	
Continue primary series (see Table 2 in the ACIP pneumococcal vaccine recommendations for complete schedule details)	
Age 2 to 5 years	
Received any complete[#] series:	
Previously received PCV-20	No additional dose of pneumococcal vaccine needed
Never received PCV-20	Administer 1 dose of PCV-20 now*
No previous PCV dose history or any incomplete[#] series:	
History of 3 PCV doses	Administer 1 dose of PCV-20 now*
History of <3 PCV doses	Administer 1 dose of PCV-20 now* AND administer additional dose of PCV-20 in ≥8 weeks
Age 6 to 18 years	
Received PCV before age 6 years but has not received PPSV23 OR received PCV13 only at or after age 6 years	
Previously received PCV-20	No additional dose of pneumococcal vaccine needed
Never received PCV-20	Administer 1 dose of PCV-20 now*
Received 1 dose of PCV13 and 1 dose of PPSV23 at or after age 6 years	
No additional dose of pneumococcal vaccine needed	

[#]Incomplete series = Not having received all doses in either the recommended series or an age-appropriate catch-up series. See Tables 8, 9, and 11 in the ACIP pneumococcal vaccine recommendations (<https://www.cdc.gov/mmwr/pdf/rr/rr5911.pdfpdf icon>) or visit <https://www2a.cdc.gov/vaccines/m/pneumo/pneumo.html> for complete schedule details; *Ensure at least 8 weeks after any prior PCV13 or PPSV23 dose.

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